

Q1

ECONOMICS

How should scarce GPU-hours be allocated for value net of carbon when teams can misreport their private value?

Q2

COMPUTATION

Can an exactly solved multi-team VCG beat first come, first served and stay auditable?

Q3

BEHAVIOR

Do people report truthfully and understand the rule? (Hugging Face Space, Section 4)

1

Strategic Reports from Teams

Teams report private value; demand and emissions are observable

Team	Demand d_i (h)	Reported r_i	CO ₂ e e_i (kg)	Score $s_i = r_i - \lambda e_i$
Team A	25	9	6.0	6.0
Team B	20	6	4.8	3.6
Team C	15	3	3.6	1.2
Team D	25	9	6.0	6.0
Team E	20	6	4.8	3.6
Team F	15	3	3.6	1.2

Information structure

- Private: project value r_i (may be misreported)
- Observable: demand d_i and estimated emissions e_i
- Carbon weight $\lambda = 0.5$; capacity $K = 100$ GPU-hours
- Total demand = 120 GPU-hours $> K$ (contested)

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Carbon-Aware VCG Allocation

Select the feasible set maximizing carbon-adjusted reported value

Optimization problem

maximize $\sum_{i \in S} (r_i - \lambda e_i)$

subject to $\sum_{i \in S} d_i \leq K$

$K = 100$ GPU-hours, $\lambda = 0.5$

r_i : reported value

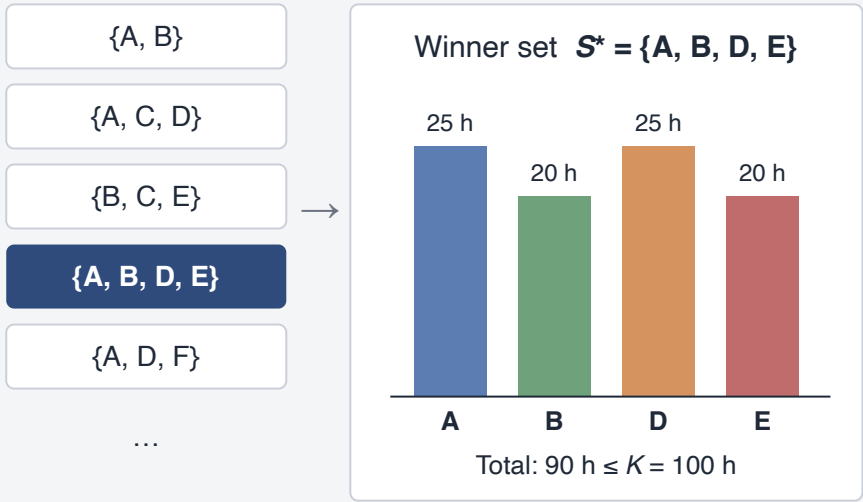
d_i : GPU-hours requested

e_i : estimated CO₂e (kg)

λ : carbon weight

K : capacity

Exhaustive search over feasible subsets



VCG payment for each winner i

$p_i = W^*_{-i} - \sum_{j \in S^* \setminus \{i\}} s_j$

p_i : VCG payment of team i

W^*_{-i} : optimal value without i

s_j : carbon-adjusted score of j

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Outcomes and Properties

Transparent allocation, payments, and carbon-aware efficiency

Resource allocation ($K = 100$ GPU-hours)



Payments (VCG)

24 priority credits each

Team A	24
Team B	24
Team D	24
Team E	24
Total	96 credits

Outcomes (selected set)

Total value $\sum r_i$	30
Carbon-adjusted score	19.2
Total CO ₂ e (kg)	21.6
kg CO ₂ e per unit value	0.72

Comparison with first-come-first-served (FCFS)

FCFS score, illustrative single order	15.6
FCFS mean score over all 720 orders	16.96
Carbon-aware VCG score	19.2

✓ No FCFS order exceeds the VCG score

Key economic properties

Truthful reporting is dominant (DSIC)	Social welfare maximization	Carbon-aware and transparent
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Incentive-compatible truthful reporting (DSIC)

Exact enumeration
720 FCFS orders / 64 VCG subsets

Lower emissions per unit of value

Practical insights for university GPU governance